

A Way of Exploring and Explaining the Mathematical World

Where do we see patterns?

- What is a good working definition of a pattern?
- How do we represent patterns with sets and functions?
- Are there different ways to represent the same patterns?
- What are the necessary and sufficient characteristics of a particular representation of a pattern?
- How do we name a pattern?
- How do we keep patterns established in reality?

How can we use patterns to find new patterns?

- In what ways can a pattern be understood as constituted of parts?
- Do we see patterns of the same kind inside a pattern?
- Do we see a pattern as being inside of a greater pattern?
- How can we combine patterns to create new patterns of the same kind?
- How can we combine a pattern inside a greater pattern with another pattern to create that greater pattern?
- What patterns can we create by either relaxing or adding to the necessary characteristics of a pattern?

How do we order patterns?

- What methods of computation or visualization help us understand patterns in relation to their own kind?
- How does one kind of pattern relate to another kind of pattern?
- When is a condition of a pattern preserved between two different patterns?
- What patterns of different representations are actually the same?
- How do the patterns inside a pattern relate to their order among other patterns?

What can we create with a pattern?

- How do we want to remember the past, express the present, or create the future?
- Who needs us to love what we love?
- What creative medium draws us into the details?
- How can a pattern be used to uniquely embody meaning in the world?